Syllabus: Mod 4
Data Management at Scale
3 credits

Mod 4 Dates
Mod 4 starts January 16, and runs through May 2, 2026.

Mod 4 Faculty
Chris Seferlis

Mod 4 Learning Facilitators
Apoorva Gupta

Your assigned Learning Facilitator is your main contact during Mod 4 for course-related matters pertaining to all course assignments, grades, and concerns directly arising within the module (i.e., course materials and supplies). To communicate with your Learning Facilitator, go to BU Virtual Campus.

Throughout Mod 4, Learning Facilitators will hold **group office hours**, and students may attend as many group office hours as they wish. Learning Facilitators are also available for individual appointments. An appointment can be requested by submitting a Help Request via the following link: BU Virtual Campus. **Please be mindful that there are a limited number of available slots for individual appointments. Therefore, use this option judiciously.**

Student Success Specialist
Annie Imperato, Learning Success Specialist.

Your Student Success Specialist will support your progression through Mod 4 and the OMDS program in general by providing valuable resources to aid your learning journey. Your Success team functions similarly to an academic coach or advisor, and this team will help you get registered and support extracurricular programming. You may reach the Student Success team at BU Virtual Campus.

Mod 4 Overview and Learning Objectives
Module 4 focuses on data management at scale. You will apply these core foundational ideas in quizzes, exams, and homework.

During Mod 4, you will:

- Describe the data science lifecycle and various types of data.
- Identify, acquire, cleanse, stage, govern, and present data using extract, transform, load (ETL)/extract, load, transform (ELT) techniques.

- Use SQL for data manipulation and relational database management.
- Discuss the significance of NoSQL databases in contemporary data management.
- Develop strategies for properly distributing data and compute across disparate systems using modern technologies such as MapReduce with Hadoop, in-memory data processing with Spark, and examples of other compute options for data processing.
- Determine the best architectures for specific big data application requirements.
- Review and understand strategies for ensuring quality data with data governance.
- Develop skills to manage data responsibly and effectively.

Mod 4 Weekly Schedule

Release Dates	Concepts	Live Sessions
Week 1: January 16	What Is Data?	Wednesday January 21 5pm ET **(delayed session due to MLK Day)**
Week 2: January 23	Data Integration and Interoperability	Monday January 26, 5pm ET
Week 3: January 30	Data Schemas and Governance	Monday February 2, 5pm ET
Week 4: February 6	Data Quality and Master Data Management	Monday February 9, 5pm ET
Week 5: February 13	Structured Data and Database Fundamentals	Tuesday February 17, 5pm ET **(delayed session due to Presidents' Day)**
Week 6: February 20	Relational Database Development and Querying	Monday February 23, 5pm ET

Week 7: February 27	Advanced Database Techniques and Data Warehousing	Monday March 2, 5pm ET
Spring Recess	**March 7 - 15**	
Week 8: March 6	Semi-Structured Data: NoSQL Databases	Monday March 16, 5pm ET **(delayed session due to recess)**
Week 9: March 20	NoSQL Databases Continued: GraphDB and Wide-Column Databases	Monday March 23, 5pm ET
Week 10: March 27	Big Data Fundamentals and Architectures	Monday March 30, 5pm ET
Week 11: April 3	Distributed Compute and Big Data Processing	Monday April 6, 5pm ET
Week 12: April 10	Real-Time Data Processing and Streaming	Monday April 13, 5pm ET
Week 13: April 17	Data Privacy, Security, and Compliance	Tuesday, April 21, 5pm ET **(delayed session due to Patriots' Day)**
Week 14: April 24	Data Visualization and Dashboards	Monday April 27, 5pm ET

Mod 4 Live Sessions

Live Sessions for **Mod 4 are offered on Mondays at 7pm ET during the term, except during holiday weeks as stated above**. Because Live Sessions are an integral part of the program and offered weekly, you are strongly encouraged to attend all Live Sessions.

If you are unable to attend a Live Session, please watch the recording. Professors consider Live Sessions essential learning, and content covered in the Live Session is eligible to be included in Module assessments.

The teaching team requests that you keep your video/web cameras on during Live Sessions (unless otherwise noted, or you may turn off your camera due to bandwidth problems).

Community Standards for Live Sessions

Live Sessions in the OMDS program are an active learning environment and include actual, real-time interactions (teaching team to student, student to student). Adherence to the Student Codes of Conduct is expected. Additionally, the program recognizes that open discourse and authentic representation of the lived experience are valuable in the learning environment. Ideas and ideologies can be framed respectfully while we all seek first to understand as well as be understood. Because OMDS encourages students to be authentic, this approach suggests that honest mistakes are possible. When this happens, learning from each other is essential. As a result, OMDS strives not to censor language or discourse during the Live Session but to demonstrate the learning opportunity as needed.

Deliverables

Mod 4 deliverables consist of quizzes, exams, and homework.

The key graded deliverables include:

Week	Type	Description / Notes
Week 2	Quiz	Covers material from Weeks 1 and 2
Week 4	Quiz	Covers material from Weeks 3 and 4
Week 5	Homework 1c	Cumulative of previous weeks' homework
Week 6	Quiz	Covers material from Weeks 5 and 6
Week 7	Exam	Covers material from Weeks 1 to 7
Week 9	Homework 2	
Week 9	Quiz	Covers Weeks 8 and 9
Week 12	Quiz	Covers Weeks 10 to 12
Week 13	Homework 3c	Cumulative of previous weeks' homework
Week 14	Exam	Covers material from Weeks 8 to 13

Weeks 1 to 14	Yellowdig	Discussion and peer engagement

Grading

There are a total of 1,000 points in Mod 4. Throughout Mod 4, you will complete activities for credit in the various gradebook assessment categories listed below. Points earned for Yellowdig engagement (post/comments) will remain reflected in Yellowdig throughout the Mod. Please make sure that you complete all your activities, including Yellowdig ones, accordingly.

We have broken the various gradebook assessment categories out by academic content, key competencies and peer engagement.

Here's how each category contributes to your overall grade:

Category	**Grading Breakdown**
Academic Content	
• Homework (Azure)	36%
• Quizzes	16%
∉ Exams	43%
Executive Skills and Peer Engagement	
• Discussion (Yellowdig)	5%

Policy Regarding Late Submissions

The OMDS program is committed to student success and strives to maintain the highest level of performance and integrity of the degree. All OMDS assignments (solo or team) are due at 11:59pm ET.

In the case of an emergency, students should submit a case via BU Virtual Campus to their assigned Learning Facilitator as soon as possible providing any relevant documentation. The Learning Facilitator will work with the Student Success team and faculty to determine if an exception to the late submission policy can be made.

Otherwise, the following late policy applies:

- All assignments are due at 11:59pm ET for full credit with a grace period of 2 hours. The grace period only applies to the original deadline, and does not apply to subsequent partial credit deadlines.
- Assignments submitted more than 2 hours and up to 5 days late will get partial credit, but penalized 10% per day for lateness (final scores rounded to the closest full point).
- Assignments submitted more than 5 days (120 hours) late will receive 0 points.
- Specifically:
  - Within 0-2 hours of original deadline: no penalty
  - Within 2-24 hours of original deadline: 90% of grade
  - Within 24-48 hours of original deadline: 80% of grade
  - Within 48-72 hours of original deadline: 70% of grade
  - Within 72-96 hours of original deadline: 60% of grade
  - Within 96-120 hours of original deadline: 50% of grade
  - More than 120 hours late: 0% of grade

Projects and exams may be subject to a different policy. Please consult the course syllabi.

Regarding incompletes, we follow BU's incomplete policy, as stated here: https://www.bu.edu/academics/policies/incomplete-coursework/

Course Materials

OMDS and BU Libraries Materials

The Blackboard **"What You Need to Know"** module will provide instructions on how to access learning materials through BU Libraries that will help you throughout Mod 4. You should plan to revisit the "What You Need to Know" folder in Blackboard frequently throughout the module.

Azure

Microsoft Azure is a cloud computing platform that will give you hands-on experience managing data at scale and applying the concepts learned throughout this course. In this module, you'll use Azure as the environment for completing your homework. Azure offers tools for data storage, processing, and analytics, enabling you to work with large datasets, experiment with different data structures, and explore various levels of data.

For this course, you will use the "Azure for Students" offer provided by Microsoft. This offer allows for a $100 credit that can be replenished once a year as long as a student

email address is being used. You will be expected to manage your budget. By adhering closely to the instructions outlined in the homework assignments, you will remain within the $100 credit limit. However, any expenses incurred beyond this allocation will be your responsibility.

Technology Advice for OMDS

For the best technical experience, students are encouraged to use a laptop or desktop computer running Mac OS X or Windows. Recommended browsers include Firefox or Chrome — access this link for more information on browser compatibility and other system requirements. In the event of technical difficulties, you can try an alternate browser or submit a help desk ticket to request help troubleshooting the problem.

While course materials are accessible from mobile devices, not all tools are optimized for these devices, and we would strongly recommend using a laptop or desktop to complete assignments and other key components of your coursework.

Requesting Accommodation

In accordance with university policy, every effort will be made to accommodate unique and special needs of students with respect to speech, hearing, vision, or other disabilities. Any student who feels they may need accommodation for a documented disability should contact BU Disability & Access Services online or by phone at 617-353-3658 for review and approval of accommodation requests. It is best to start this process as early as possible.

Sexual Misconduct/Title IX Policy

Boston University is committed to fostering a safe learning environment for all members of its community and preventing sexual misconduct. All forms of sexual misconduct, including rape, acquaintance rape, sexual assault, domestic and dating violence, stalking, and sexual harassment, are violations of Boston University's policies, whether they happen on campus or off campus.

Title IX of the Education Amendments Act of 1972 is a federal civil rights law that prohibits sex-based discrimination in federally funded education programs and activities. This law makes it clear that violence and harassment based on sex and gender are civil rights offenses subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories, such as race, national origin, etc. If you or someone you know has been harassed or assaulted, you can find the appropriate resources at Sexual Misconduct Safety.

Academic Integrity

Students are expected to do their own work, not cheat, and must abide by Boston University's Academic Conduct Code.

"Boston University is a community of learning grounded in shared values, rights, and responsibilities. One critical component of our learning community is academic integrity, that is a commitment, even in the face of adversity, to the fundamental values of honesty, trust, fairness, respect, responsibility, and courage in representing truthfully one's knowledge and ideas as one's own. Our community can only be maintained if all students represent honestly their own knowledge and ideas, believe that their academic competence is being judged fairly based on their knowledge and ideas, and have confidence that they will not be put at a disadvantage because of someone else's dishonesty. Promoting and protecting academic integrity is the responsibility of every member of the campus community because our goals of teaching, learning, and research can only be accomplished in environments in which ethical standards are upheld.

"Academic dishonesty undermines the bonds of trust and honesty between members of the community and defrauds those who depend upon our knowledge and integrity. Students, faculty, and staff of Boston University support this Academic Conduct Code for the purposes of providing an atmosphere of mutual trust and respect, fostering ethical behavior, and cultivating lifelong professional conduct."

Solo assignments must be completed with solo effort, and any and all work must have proper citation for all OMDS assignments. Team assignments may only be shared with your designated team. The OMDS program utilizes Turnitin (a plagiarism detection software) to assess submitted assignments. Any student believed to be in violation of the Academic Conduct Code for any reason will be subject to the university process as outlined at Academic Conduct Code.

Students should learn how to use artificial intelligence (AI) text generators and other AI-based assistive resources (collectively, AI tools) to enhance rather than damage their developing abilities as writers, coders, communicators, and thinkers. The CDS Generative AI Assistance (GAIA) Policy provides guidance on how to use AI tools safely and responsibly while adhering to the Academic Conduct Code.

Community Code of Conduct

As members of the OMDS community, each of us has a collective and individual responsibility to conduct ourselves in accordance with Boston University's core values:

1. Collaborate with each other and the teaching team (i.e., faculty and Learning Facilitator team) to foster a culture of inclusion.
2. Treat each other with dignity, fairness, and respect.
3. Communicate using a professional tone and clarity to model the behavior of a business leader.
4. Debate respectfully in online forums.
5. Give feedback directly to others respectfully and constructively.

Violations of the community code of conduct pose a challenge to the overall student code of conduct and may be subject to administrative response and consequence. Students are valued members of the OMDS community and can be assured that OMDS faculty and staff also adhere to the same community code of conduct.

*The OMDS program reserves the right to modify the requirements of the Mod with adequate notice to students and within the scope of authority and responsibility entrusted by Boston University.